Portfolio Deliverable 1 | Target Discovery Package
Multi-Omics Target Discovery & Structural Prioritization for Parkinson's Disease

SECTION 1 — DISEASE AND QUESTION

Disease Context: Parkinson’s disease (PD) is a progressive neurodegenerative disorder characterized by the selective loss of dopaminergic neurons in the *substantia nigra pars compacta*. The primary pathophysiology involves pathological protein aggregation (predominantly -synuclein) and progressive cell death, leading to severe motor and non-motor deficits.

Core Question: Which genes demonstrate convergent, multi-layered evidence across genomics, transcriptomics, and proteomics, and among those convergent hits, which represent druggable and structurally tractable targets for therapeutic intervention?

What a Good Target Looks Like Here:

A promising therapeutic target for Parkinson's disease requires:

1. **Robust Genetic Support:** Statistically significant risk association in human GWAS data.
2. **Tissue-Specific Alteration:** Observable, concordant dysregulation at the transcript and/or protein level in affected human *substantia nigra* tissue.
3. **Plausible Drug Modality:** High structural/sequence-based Druggability compatibility for small molecules, biologics, or targeted protein degradation (PROTACs).
4. **Confident Structural Tractability:** High-confidence 3D structural models featuring accessible, well-defined, and stable binding pockets or catalytic clefts.

SECTION 2 — MULTI-OMICS INTEGRATION (FIGURES 1 AND 2)

Integrated Datasets and Cohort Matching

To identify high-confidence PD targets, evidence was integrated across three distinct omics layers:

- **Genomics:** GWAS Catalog (MONDO_0005180, Parkinson’s disease) comprising 698 risk associations mapped to **393 unique genes**.
- **Transcriptomics:** GSE7621 dataset (Lesnick et al., 2007) of post-mortem *substantia nigra* tissue (16 PD vs. 9 controls) profiling **20,799 genes**.
- **Proteomics:** Licker et al. (2014) post-mortem *substantia nigra* dataset (PXD000427; 5 PD vs. 8 controls) identifying **1,779 proteins**.

Access & Matching Caveat: All datasets are open access. However, because the transcriptomic, proteomic, and genomic cohorts are completely unmatched (derived from distinct patient groups and clinical studies), patient-level multi-omics integration was impossible, restricting the analysis strictly to gene-level overlap.

Cross-Layer Overlap and Concordance Analysis

Intersection Metric	Gene Count
Genomics total	393
Transcriptomics total	20,799
Proteomics total	1,779
RNA Protein overlap	1,684
Three-layer intersection (Genomics RNA Protein)	**34**

Across the 34 convergent genes identified in all three layers, expression concordance between RNA log2FC and protein log2FC was systematically assessed:

- **Concordance Rate: 16 / 34 (47.1%)**.
- **Correlation:** $\rho = -0.136$
- **Interpretation:** There is **no detectable agreement** between transcriptomic and proteomic fold-change directions in the affected *substantia nigra* tissue.

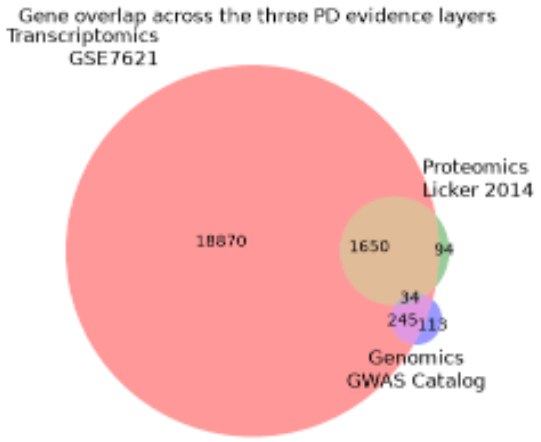


Figure 1. Gene overlap across the three Parkinson's disease evidence layers (GSE7621 transcriptomics, Licker et al. 2014 proteomics, GWAS Catalog genomics). Circle areas are not to scale; numbers are gene counts.

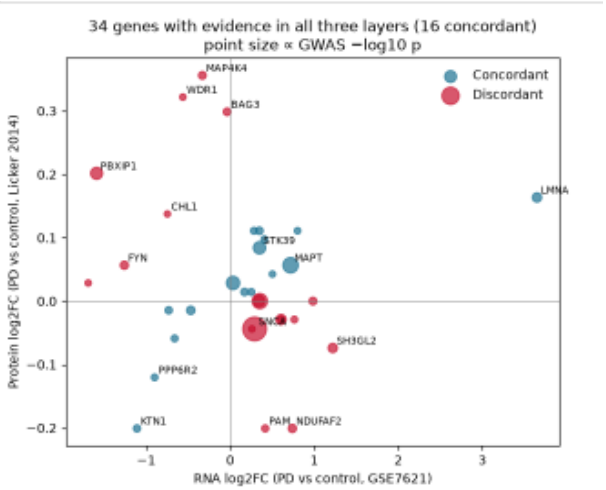


Figure 2. RNA versus protein log2 fold change (PD vs control) for the 34 genes present in all three layers. Blue = concordant (16 genes), red = discordant (18 genes); point size is proportional to GWAS −log10 p

Key Biological Hits and Locus Caveats

Known PD Genes Recovered:

- *SNCA* (Rank #3): Recovered in all three layers, driven predominantly by overwhelming GWAS signal (-10p=169.4).
- *MAPT* (Rank #9): Recovered across all three layers, strongly driven by GWAS (-10p=68.0).

Known PD Genes Lost:

- *LRRK2* and *GBA1*: Present in the GWAS genomic layer but missed in the target shortlist because both were absent from the restricted ~1,800-protein proteomic coverage set.

- **Non-Obvious Hit (*LMNA* - Rank #1):**
- *LMNA* exhibited a massive RNA log2FC of +3.65, but **no RNA probe passed FDR correction** (minimum adjusted p-value = 0.538); its strongest probe (244225_x_at) is an _x_at probe that may cross-hybridize, and LMNA's other probes show smaller, mixed changes.

Discordant Hit (*PBXIP1* - Rank #2):

- *PBXIP1* displayed strong directional discordance: RNA log2FC of −1.59 (non-significant, adj p=0.771) versus protein log2FC of +0.20 (statistically significant), coupled with a strong GWAS p-value (-10p=38.7).

Locus Mapping Caveat:

- Both *LMNA* and *PBXIP1* map to the **chr1q21–q22 locus**, in close physical proximity to *GBA1*. Given that GWAS signals were assigned via nearest-gene mapping, their high genomic scores likely represent synthetic association artifacts driven by the causal *GBA1* locus.

SECTION 3 — DRUGGABILITY TRIAGE

Model Architecture, Validation, and Leakage Guard

Model: Logistic regression prior trained on 960-dimensional ESM C 300M protein embeddings using Open Targets clinical-precedence labels (sm_clinical and ab_clinical).

Validation Performance:

- *Family-split CV (Generalizability test):* AUROC = **0.754**, Average Precision (AP) = **0.555** (vs. random baseline AP = 0.334).
- *Random-split CV:* AUROC = 0.848, AP = 0.731.

Data Leakage Guard: Two of my targets that were also in the training panel (SNCA and SH3GL2) were removed before fitting, so no target was predicted using its own label.

Inverse Relation and Weighting Scheme

A key finding from Module 3 was a strong inverse correlation between biological disease evidence (Module 2) and predicted drug-likeness (Module 3):

- **Correlation:** =-0.618.
- *Interpretation:* The genes with the strongest genetic/disease evidence look the least historically "drug-like" to sequence-only language models.
- **Composite Weighting:** A composite score weighting of w=0.6 (Disease Evidence) / 0.4 (Druggability) was established. The top two targets (MAP4K4, PBXIP1) were unchanged across w = 0.4 to 0.8; only the third pick depended on the weight.

Re-Ranked Candidates and Model Blind Spots

Re-Ranked Top 3 Candidates:

1. **MAP4K4:** psm=0.665, pab=0.000 (Re-ranked **#1**).
2. **PBXIP1:** psm=0.001, pab=0.037 (Re-ranked **#2**).
3. **SNCA:** psm=0.023, pab=0.001 (Re-ranked **#3**).

- **Most Drug-like Overall: FYN** (psm=0.96, sequence neighbors *SRC* 0.99, *LYN* 0.97) but placed low in overall priority (14) due to weak initial disease evidence.
- **Model Blind Spot (*SNCA*):** Despite having active clinical precedence (ab_clinical = True), ESM C predicted a near-zero antibody probability (pab=0.001) because its nearest clinically precedented neighbors in embedding space (PSMD4, TNNT3, RPL7A) had cosine similarities of only 0.84–0.85, the weakest matches of all 15 targets.
- **Candidates Carried Forward: MAP4K4** and **SNCA**. PBXIP1 was swapped for SNCA because of SNCA's central role in PD pathology, accepting that its disorder makes pocket assessment harder (Module 3 addendum). Later Module 2 also weakened PBXIP1: its GWAS signal lies near GBA1, and its RNA change is not FDR-significant.

SECTION 4 — STRUCTURAL ASSESSMENT (FIGURE 3)

Metric / Assessment Feature	**MAP4K4 (UniProt: O95819)**	**SNCA (UniProt: P37840)**
Mean pLDDT	65.5	75.2
% Residues pLDDT < 50	36.6%	12.9%
Domain Architecture	3 domains; inter-domain PAE 25–29 Å (domain arrangement uncertain)	1 continuous domain
Experimental Reference	PDB **4U43** Chain A (Kinase domain + inhibitor 3D8, 2.18 Å)	PDB **1XQ8** Chain A (Micelle-bound NMR structure)
RMSD (Confident Residues)	**1.19 Å** over 259 residues (kinase domain)	**32.45 Å** whole-chain; helix 3–37: 1.36 Å, helix 45–92: 3.77 Å
Ligand Site Metrics	15 inhibitor-contact residues: mean pLDDT 87.1, mean PAE 2.9 Å	N/A (no ligand-bound catalytic site)
Pocket Identification	fpocket merged catalytic site into Pocket 122 (87% coverage, score 0.002, **rejected**); best kept Pocket 60 (score 0.103, 0% overlap)	Pocket 1 score 0.949, pLDDT 89.4, PAE 7.4 Å (located on one flat helix face; lacks 3D enclosure)

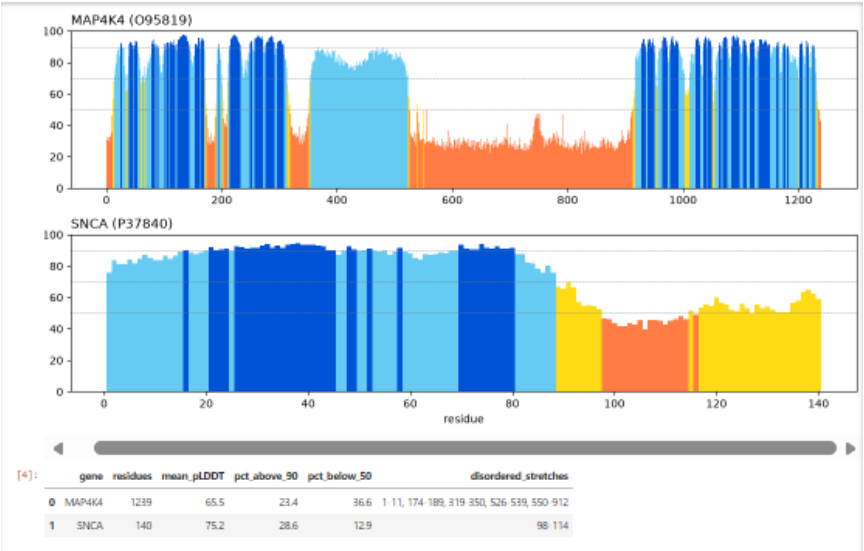


Figure 3. (A) AlphaFold DB models of MAP4K4 (O95819) and SNCA (P37840), colored by pLDDT. (B) Per-residue pLDDT; low-confidence stretches (pLDDT < 50): MAP4K4 1–11, 174–189, 319–350, 526–539, 550–912; SNCA 98–114. Models from AlphaFold DB (Varadi et al., 2022).

Structural & Pocket Claim Interpretations

- **MAP4K4 Pocket Claim:** fpocket failed to cleanly isolate the true catalytic site on full-length AlphaFold predictions because it merged low-confidence surface loops (residues 172–191) and distant C-terminal segments into Pocket 122. However, superposition against 4U43 gives a 1.19 Å RMSD over 259 kinase-domain residues. The catalytic pocket residues themselves exhibit high local confidence (pLDDT 87.1, PAE 2.9 Å), supporting a well-modelled inhibitor (3D8) site within the kinase domain. The rest of the protein is less certain: 36.6% of residues have pLDDT < 50, including a long-disordered region (550–912), and the arrangement of the three domains is uncertain.
- **SNCA Pocket Claim:** Although fpocket assigned Pocket 1 a high score (0.949), structural inspection reveals this "pocket" is merely an exposed face along a single amphipathic helix that **lacks 3D enclosure**. Because monomeric SNCA is intrinsically disordered in solution and requires a lipid/micelle environment to form stable helices, this predicted surface groove is unlikely to support stable small-molecule binding.

Exact Druggability Calls

- **MAP4K4: Druggable with minor structural caveats**.
- **SNCA: Consider another modality (biologics / degradation / nucleic acid)**.

SECTION 5 — PRIORITIZED TARGET PACKAGE (TABLE 1)

Target Prioritization Matrix

Gene	Module 2 rank	GWAS −log10 p	RNA / protein log2FC	Module 3 p_sm / p_ab (rank)	Module 4 call	Final priority
MAP4K4	4	12.1	−0.33 (ns) / +0.36 (sig)	0.665 / 0.000 (1)	Structure supports a small-molecule pocket	Tier 1
SNCA	3	169.4	+0.29 (ns) / −0.04 (ns)	0.023 / 0.001 (3)	No confident pocket: consider another modality	Tier 2 (alternative modality)
PBXIP1	2	38.7	−1.59 (ns) / +0.20 (sig)	0.001 / 0.037 (2)	Not assessed	Deprioritized
LMNA	1	20.7	+3.65 (ns) / +0.16 (sig)	0.004 / 0.002 (4)	Not assessed	Deprioritized

Table 1. Prioritized target shortlist integrating multi-omics evidence (Module 2), sequence-based tractability (Module 3), and structural assessment (Module 4). ns = not FDR-significant (RNA) or not flagged (protein); sig = flagged significant by Licker et al.

Documented Limitations

1. **Unmatched Cohorts:** Transcriptomic, proteomic, and genomic datasets originated from distinct patient cohorts, precluding individual-level multi-omics co-expression analysis.
2. **Lack of Transcriptomic FDR Significance:** No individual transcript passed strict false discovery rate (FDR) thresholds in the GSE7621 dataset.
3. **Small Sample Sizes:** Limited statistical power in underlying tissue datasets (RNA: 16 PD / 9 controls; Proteomics: 5 PD / 8 controls).
4. **GWAS Mapping & 1q22 Ambiguity:** Nearest-gene mapping in GWAS Catalog introduces synthetic loci hits (*LMNA*, *PBXIP1*) near strong causal loci like *GBA1*.
5. **Hyphen-Split Processing Bug:** A string parsing bug in early pipeline scripts was identified, though verification confirmed it did not impact the top 15 candidates.
6. **Sequence-Only Druggability Blind Spots:** PLM priors fail on non-traditional drug targets like intrinsically disordered proteins (*SNCA*).
7. **In Silico Structural Modeling Limits:** AlphaFold monomer predictions lack ligands, lipid/membrane environments, and obligate binding partners.
8. **fpocket Pocket Definition:** on full-length models fpocket can merge a real ligand site with neighboring low-confidence residues (MAP4K4 pocket 122), so pocket scores must be checked against known ligand sites.

Proposed Validation Experiment for MAP4K4

Biochemical Activity & Cellular Target Engagement Assay: To validate **MAP4K4** as the top small-molecule target, conduct a two-step wet-lab experiment:

1. *In Vitro Kinase Inhibition:* Express human MAP4K4 kinase domain (residues 1–320) and run a microfluidic radiometric ATP-competitive kinase assay against a focused library of MAP4K4 inhibitors (e.g., 3D8 analogs) to measure IC50 binding kinetics.
2. *Cellular Engagement in Human iPSC Dopaminergic Neurons:* Treat patient-derived iPSC dopaminergic neurons with top inhibitor hits and assess target engagement via CETSA (Cellular Thermal Shift Assay) alongside Western blot quantification of phosphorylated downstream substrates (e.g., p-Ezrin/Radixin/Moesin).

Go/No-Go Mind-Changing Result:

If nanomolar small-molecule inhibition of MAP4K4 fails to attenuate autophagic stress or protect against -synuclein-induced loss of dopaminergic neurons in human iPSC models, MAP4K4 will be discarded as a therapeutic target for Parkinson's disease.

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